

Software Tester (QA) Assessment

Test Case Design & Bug Identification — Task Management Application

1. Test Scenarios

ID	Scenario	Description
TS-01	Registration	Verify new user can register with valid details; verify duplicate/invalid inputs are rejected.
TS-02	Registration (Negative/Edge)	Existing email, invalid email format, weak password, empty fields, password mismatch, max-length & special-character inputs.
TS-03	Login	Valid login, invalid credentials, empty fields, password masking, account lockout after repeated failures, session expiry.
TS-04	Create Task	Create task with valid data; reject empty title; handle max-length, special characters, whitespace-only input.
TS-05	Edit Task	Edit and save task, cancel edit, edit another user's task (security), concurrent edit handling.
TS-06	View Task List	List shows correct tasks sorted/filtered correctly; empty list message; large list pagination/performance.
TS-07	Delete Task	Delete with confirmation, verify removal from DB, delete already-deleted task, cancel delete.
TS-08	Input Validation & Error Handling	XSS/SQL injection attempts, invalid date formats, server/network error messages, no stack traces leaked to UI.

2. Test Cases

2.1 Registration

TC ID	Test Case	Precondition	Test Steps	Test Data	Expected Result	Type/Priority
TC-R01	Register with valid details	Registration page open	1. Enter name, email, password, confirm password 2. Click Register	name=Ram, email=ram@test.com, pwd=Pass@123	Account created; user redirected to login/dashboard; success message shown	Positive / High
TC-R02	Register with existing email	User already registered	1. Enter already-registered email 2. Click Register	email=ram@test.com	Error: 'Email already registered'; no duplicate account created	Negative / High
TC-R03	Invalid email format	Registration page open	1. Enter 'ram@abc' (no domain) 2. Submit	email=ram@abc	Inline validation error; form not submitted	Negative / Medium
TC-R04	Weak password	Registration page open	1. Enter password '12345' 2. Submit	pwd=12345	Error: password must meet policy (min length, upper, digit, special)	Negative / High
TC-R05	Empty mandatory fields	Registration page open	1. Leave fields blank 2. Submit	all empty	Required-field errors shown; no submission	Negative / High
TC-R06	Password & confirm mismatch	Registration page open	1. Enter two different passwords 2. Submit	pwd=Pass@123 / confirm=Pass@124	Error: passwords do not match	Negative / Medium
TC-R07	Max-length input (edge)	Registration page open	1. Enter 255-char name/email 2. Submit	255-char strings	Handled gracefully: accepted within limit or clear error; no crash/UI break	Edge / Low
TC-R08	XSS/SQLi in registration fields	Registration page open	1. Enter alert(1) / ' OR 1=1-- in name field 2. Submit	script & SQLi payloads	Input sanitized/encoded; no script execution or DB error	Negative / Critical

2.2 Login

TC ID	Test Case	Precondition	Test Steps	Test Data	Expected Result	Type/Priority
TC-L01	Login with valid credentials	Registered user exists	1. Open login 2. Enter valid email/password 3. Click Login	ram@test.com / Pass@123	User logged in; redirected to task list; session created	Positive / High
TC-L02	Wrong password	Registered user exists	1. Enter valid email, wrong password 2. Login	ram@test.com / Wrong@1	Error: invalid credentials; no login	Negative / High
TC-L03	Unregistered email	—	1. Enter random email 2. Login	nobody@test.com / Pass@123	Generic error shown (no user enumeration)	Negative / High

TC ID	Test Case	Precondition	Test Steps	Test Data	Expected Result	Type/Priority
TC-L04	Empty fields	Login page open	1. Leave blank 2. Click Login	empty	Validation errors; no request sent	Negative / Medium
TC-L05	Password masking	Login page open	1. Type password in field	Pass@123	Characters shown as dots/asterisks	Positive / Medium
TC-L06	Account lockout (edge)	Registered user exists	1. Enter wrong password 5 times	wrong pwd x5	Account temporarily locked / CAPTCHA shown; prevents brute force	Edge / Critical
TC-L07	Back button after logout	User logged in	1. Login 2. Logout 3. Press browser Back	—	User NOT able to see cached task page; redirected to login	Negative / High
TC-L08	SQL injection in login	Login page open	1. Enter ' OR 1=1-- in email, any pwd 2. Login	' OR 1=1--	Login rejected; parameterized query used; no error page	Negative / Critical

2.3 Task CRUD Operations

TC ID	Test Case	Precondition	Test Steps	Test Data	Expected Result	Type/Priority
TC-C01	Create task with valid data	User logged in	1. Open create task 2. Enter title & description 3. Save	title='Buy groceries'	Task saved; appears in list; success toast	Positive / High
TC-C02	Create task with empty title	User logged in	1. Open create task 2. Leave title blank 3. Save	title=""	Error: title is required; task not saved	Negative / High
TC-C03	Whitespace-only title (edge)	User logged in	1. Enter ' ' as title 2. Save	title=' '	Trimmed and rejected OR saved trimmed; consistent behaviour	Edge / Medium
TC-C04	Create with max-length/special chars	User logged in	1. Enter 500-char title with emojis/symbols 2. Save	long + special chars	No truncation/corruption; UI not broken; or clear limit enforced	Edge / Medium
TC-E01	Edit task and save	Task exists	1. Open task 2. Edit title 3. Save	new title='Buy milk'	Updated value shown in list & DB	Positive / High
TC-E02	Cancel edit	Task exists	1. Open task 2. Change text 3. Cancel	—	Original data unchanged	Positive / Medium
TC-E03	Edit another user's task (security)	2 users, User A's task exists	1. Login as User B 2. Try to open/edit User A's task via URL	Task ID of A	Access denied (403); task cannot be modified	Negative / Critical
TC-E04	Concurrent edit (edge)	Task exists, 2 sessions open	1. User A & B edit same task simultaneously 2. Both save	different values	Last-writer-wins OR conflict warning; no data corruption	Edge / Major
TC-V01	View task list	User has tasks	1. Login 2. Open task list	—	All tasks of logged-in user shown; correct order; no other user's tasks	Positive / High
TC-V02	Empty list (edge)	New user, no tasks	1. Login 2. Open task list	—	Friendly empty-state message; no errors	Edge / Low
TC-V03	Large list (edge)	1000+ tasks exist	1. Open task list / scroll	—	Pagination or lazy loading; page loads fast; no memory/UI issues	Edge / Major
TC-D01	Delete task with confirmation	Task exists	1. Click Delete 2. Confirm	—	Task removed from UI and DB; success message	Positive / High
TC-D02	Cancel delete	Task exists	1. Click Delete 2. Cancel on dialog	—	Task remains intact	Positive / Medium
TC-D03	Double-click delete / refresh check	Task exists	1. Delete task 2. Refresh page	—	Task stays deleted (not a soft UI-only removal)	Negative / Major
TC-D04	Delete already-deleted task (edge)	Task deleted in another tab	1. Try deleting same task again	same task ID	Graceful message: task not found; no crash	Edge / Medium

2.4 Input Validation & Error Handling

TC ID	Test Case	Precondition	Test Steps	Test Data	Expected Result	Type/Priority
TC-VL 01	XSS in task title	User logged in	1. Create task with alert('x') 2. Open list	script payload	Payload stored encoded; no popup on list view; no script execution	Negative / Critical
TC-VL 02	SQL injection in task search/filter	User logged in, tasks exist	1. Search/filter with ' OR 1=1--	' OR 1=1--	No error; treated as literal text; parameterized queries used	Negative / Critical
TC-VL 03	Invalid date format	Task has due-date field	1. Enter 32/13/2025 or text as date	invalid date	Clear validation error; no crash/garbage save	Negative / Medium
TC-VL 04	Network/server failure handling	User logged in	1. Disconnect network 2. Save task	—	User-friendly error; no data loss; graceful retry, no stack trace shown	Negative / Major
TC-VL 05	Error message quality	User logged in	1. Trigger 404/500 by bad request	—	Generic friendly message; NO internal paths/SQL/stack traces leaked	Negative / Major

3. Bug Identification — Potential Bugs / Risk Areas (without execution)

Bug ID	Bug / Risk Description	Severity
BG-01	Passwords may be stored in plain text or with weak hashing (MD5/SHA1) if security review was skipped — risk of full credential leak on DB breach.	Critical
BG-02	No account lockout/CAPTCHA after repeated failed logins — enables brute-force and credential-stuffing attacks on the login endpoint.	Critical
BG-03	Session not invalidated on logout or session ID not rotated after login — session fixation/hijack risk; back-button may reveal cached pages.	Critical
BG-04	Stored XSS via task title/description — malicious script executes for every user viewing the task list (session theft, defacement).	Critical
BG-05	SQL injection possible in login or task search/filter if queries are string-concatenated instead of parameterized — full DB read/write risk.	Critical
BG-06	Deleted task still visible until manual refresh — frontend not synced with backend state; stale cache leads to user confusion and wrong counts.	Major
BG-07	Double-click on Save/Create/Delete fires duplicate requests — duplicate tasks created or delete executed twice; missing request debouncing/idempotency.	Major
BG-08	No max-length validation on task title — overlong input breaks UI layout, causes DB truncation errors, or app crashes on save.	Major
BG-09	Error responses may expose stack traces, SQL queries or internal paths — information leakage useful to attackers for further exploitation.	Major
BG-10	Timezone/date handling inconsistency — due dates stored in one timezone and displayed in another, showing wrong dates to users.	Minor

Approach Note

Since the application has no test documentation and is near production release, I prioritized: (1) security risks (auth, session, injection, XSS) as Critical, since these directly impact release; (2) data-integrity risks (delete/sync, duplicate submits) as Major; and (3) UX/consistency issues as Minor. Test cases cover positive, negative and edge paths, with an assumption that standard QA practices like input validation, parameterized queries and proper session management should hold, and each 'potential bug' above is a risk area to verify first.